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$$\int \sqrt{x+6} dx$$

$$u = x + 6$$

$$\Rightarrow \int \sqrt{x+6} dx = \int \sqrt{u} du$$

$$\Rightarrow \int \sqrt{u} = u^{\frac{1}{2}} \Rightarrow \int \sqrt{u^{\frac{1}{2}}} du$$

$$\Rightarrow \int u^n du = \frac{u^{n+1}}{n+1} + C \Rightarrow \frac{u^{\frac{3}{2}}}{\frac{3}{2}} = \frac{2}{3} u^{\frac{3}{2}}$$

$$\Rightarrow \frac{2}{3} (x+6)^{\frac{3}{2}} + C$$

References